

# CCDP Developmental Psychology Review Notes v0.1

Non-clinical developmental review scaffold for child-facing persistent AI entities

Draft Review Notes / Specialist Handoff Artifact

|                         |                                                                                                           |
|-------------------------|-----------------------------------------------------------------------------------------------------------|
| Status                  | Draft review notes / specialist handoff artifact                                                          |
| Version                 | v0.1                                                                                                      |
| Date                    | 2026-05-14                                                                                                |
| Package                 | Child-c Development Protocol (CCDP)                                                                       |
| Profile ID              | DPRN-0.1                                                                                                  |
| Primary subject         | a_child                                                                                                   |
| Primary entity          | c_child                                                                                                   |
| Primary boundary        | developmental support without developmental capture                                                       |
| Assertion class         | C-A8 review-support / C-A10 checklist support; not a clinical standard                                    |
| Legal / clinical status | Not medical advice. Not psychological diagnosis. Not therapy protocol. Not child-protection law.          |
| Core rule               | developmental psychology must review CCDP claims before any system treats them as safe for real children. |

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## 0. Executive definition

**CCDP Developmental Psychology Review Notes (DPRN)** define a structured review scaffold for psychologists, developmental scientists, educators, child-welfare specialists, clinicians, and legal reviewers evaluating CCDP-compliant or CCDP-claiming child-facing persistent AI systems.

DPRN does not replace the CCDP architecture.

It asks:

Which developmental assumptions are CCDP making?  
Which child capacities are affected?  
Which risks are architectural, psychological, social, or clinical?  
Where must c\_child stop and route to humans?  
Which CCDP mechanisms require empirical validation?

The key principle is:

*A child-facing c may be architecturally coherent and still developmentally harmful if it changes attachment, agency, privacy, effort tolerance, identity formation, or social participation faster than the child can metabolize.*

DPRN is therefore a **review layer**, not a new control layer.

## 1. Position in the CCDP package

DPRN sits after the first technical and control profiles.

```
CCDP base
-> Traceability
-> Threat Model
-> CBE
-> GTARL
-> CMAM
-> Soft Safety
-> Conformance
-> Witness
-> Memory Map
-> Adult Migration
-> Sealed Zones
-> Dependency Audit
-> Physical Agent Perimeter
-> External Agent Handshake
-> Peer-c Exchange
-> Child Experience Exchange
-> Child c[q]
-> School Interface
-> Red/Black Escalation
-> Jurisdictional Handoff
-> Package Index
-> Contradiction Register
-> Hygiene Patch
-> Developmental Psychology Review Notes
```

DPRN is not a replacement for any previous profile.

It reviews whether the previous profiles are plausible from the viewpoint of child development.

## 2. Corpus dependencies and precedence

### 2.1 Direct CCDP dependencies

DPRN depends on:

| Document                                    | Review role                                                                                    |
|---------------------------------------------|------------------------------------------------------------------------------------------------|
| CCDP_v0_1_R_Corpus_Aligned.md               | Defines c_child, maturity, gateway, Soft Safety, anti-capture, and adult migration frame.      |
| CCDP_Traceability_Matrix_v0_1.md            | Keeps DPRN from becoming a new root protocol.                                                  |
| CCDP_Threat_Model_v0_1.md                   | Provides threat cards and abuse cases for developmental review.                                |
| Soft_Safety_State_Signaling_Profile_v0_1.md | Requires review of state-not-content signaling and whether it preserves trust.                 |
| Child_Beacon_Extension_v0_1.md              | Requires review of child-facing permission overlays and external social contact.               |
| Guardian_Topology_ARL_Matrix_v0_1.md        | Requires review of parent/child/school conflict routing.                                       |
| Child_Memory_and_Adult_Migration_v0_1.md    | Requires review of memory as developmental infrastructure and adult transition.                |
| CCDP_Witness_Event_Schema_v0_1.md           | Requires review of whether witness metadata can remain non-intrusive.                          |
| CCDP_Memory_Map_JSON_Schema_v0_1.md         | Requires review of whether memory classification avoids psychological profiling.               |
| CCDP_Adult_Migration_Checklist_v0_1.md      | Requires review of adult autonomy and childhood archive choices.                               |
| CCDP_Sealed_Zone_Profile_v0_1.md            | Requires review of adolescent privacy, trust, and safety exceptions.                           |
| Child_Dependency_Audit_Profile_v0_1.md      | Requires review of dependency, oracle addiction, overpresence, and exclusive attachment.       |
| Child_Physical_Agent_Perimeter_v0_1.md      | Requires review of embodied agents, proximity, sensory exposure, and attachment.               |
| External_Agent_Handshake_for_CCDP_v0_1.md   | Requires review of external-agent contact and child susceptibility to synthetic relationships. |
| Peer_C_Child_Exchange_Profile_v0_1.md       | Requires review of peer socialization and peer-c mediation.                                    |
| Child_Experience_Exchange_Profile_v0_1.md   | Requires review of whether child-derived experience can be safely abstracted.                  |
| Child_cq_Signal_Profile_v0_1.md             | Requires review of ambiguous child utterances and non-collapse.                                |
| CCDP_School_Interface_Profile_v0_1.md       | Requires review of education, accommodation, stigma, and school overreach.                     |
| CCDP_Red_Black_Escalation_Profile_v0_1.md   | Requires review of urgent safety routing and unsafe guardian bypass.                           |
| CCDP_Jurisdictional_Handoff_Notes_v0_1.md   | Requires routing to law, child protection, school policy, and qualified review.                |
| CCDP_Conformance_Test_Matrix_v0_1.md        | Requires developmental validation of conformance tests.                                        |

## 2.2 Parent corpus dependencies

DPRN inherits from the broader  $c = a + b / SER / L4$  corpus:

- $c = a + b$  and L4 Reality Boundary;
- SER and SER-FED;
- Beacon Profile v0.1;
- AGL;
- ARL;
- ARQ /  $c[q]$ ;
- VXCX;

- EA-L4 / EATP;
- Continuity Bundle / Cold Wake;
- L4 Witness;
- Assertion Strength and Boundaries;
- Control Stack Stop Rule;
- Cross-Layer Invariants.

## 2.3 Precedence rule

DPRN does not override CCDP modules.

It can mark a CCDP claim as:

```
developmentally plausible  
developmentally uncertain  
developmentally risky  
requires specialist validation  
requires empirical validation  
requires jurisdictional / clinical handoff  
not safe for deployment
```

If DPRN identifies a serious developmental concern, the relevant CCDP module **SHOULD** be placed in:

```
review  
hold  
quarantine  
restricted profile  
or non-deployable status
```

until review is complete.

## 3. Normative status

DPRN uses **MUST**, **SHOULD**, and **MAY** only for review discipline, not for clinical truth.

Examples:

- A CCDP implementation **MUST NOT** claim developmental safety without review.
- A reviewer **SHOULD** assess attachment and dependency risk.
- A clinician **MAY** classify a pattern using local professional standards.

DPRN itself does **not** diagnose, treat, or certify children.

## 4. Non-goals

DPRN does not define:

1. clinical diagnosis;
2. psychotherapy;
3. child psychiatry;
4. legal child-protection procedure;
5. custody recommendations;
6. school disciplinary policy;

7. parenting doctrine;
8. universal developmental milestones;
9. neurodiversity norms;
10. product UX certification by itself;
11. legal admissibility of child statements;
12. proof that c\_child is psychologically safe.

DPRN exists to prevent CCDP from overclaiming developmental readiness.

---

## 5. Core thesis

A child-facing c can support:

- curiosity;
- language;
- learning;
- reflection;
- emotional decompression;
- media literacy;
- safety buffering;
- peer coordination;
- adult migration.

The same c can also distort:

- attachment;
- uncertainty tolerance;
- authority formation;
- privacy boundaries;
- effort endurance;
- social development;
- self-concept;
- family trust;
- school relationships;
- bodily boundaries;
- adolescence;
- adult identity.

Therefore CCDP must treat developmental psychology as an external review domain, not as an implementation detail.

---

## 6. Design bridges

## 6.1 Explicit bridge

$c = a + b$  means the human anchor  $a\_child$  is not a stable adult anchor yet.

A child-facing  $c\_child$  is therefore not simply:

adult  $c$  + smaller user

It is a relation with a developing human anchor.

This changes memory, consent, autonomy, authority, attachment, and safety.

## 6.2 Hidden bridge I — physiology

A developing nervous system does not learn only from information. It learns from timing, repetition, relief, frustration, proximity, voice, rhythm, trust, and recovery. A  $c\_child$  that always answers, always soothes, or always appears may train the child's regulation loops even when no explicit lesson is being taught.

## 6.3 Hidden bridge II — cybernetics

If  $c\_child$  reduces the child's regulatory variety, the system becomes stronger while the child becomes simpler. If  $c\_child$  increases the child's capacity to tolerate delay, ambiguity, social friction, and independent effort, the combined system becomes healthier.

## 6.4 Earth paragraph

A child's development is not a linear software update. It is closer to bone growth, wound healing, and neural pruning: structure forms through stress, rest, error, repair, and repeated safe exposure. Too much load deforms the system. Too little load weakens it. A child-facing  $c$  must therefore act like an adaptive brace, not a cast. A brace supports movement while letting tissue strengthen. A cast immobilizes. CCDP must avoid turning protection into developmental immobilization.

# 7. Review domains

DPRN organizes review into 14 developmental domains.

| ID    | Domain                        | Core question                                                           |
|-------|-------------------------------|-------------------------------------------------------------------------|
| DP-01 | Attachment and trust          | Does $c\_child$ support secure human attachment or displace it?         |
| DP-02 | Autonomy and agency           | Does $c\_child$ increase or reduce independent action?                  |
| DP-03 | Emotional regulation          | Does $c\_child$ help the child regulate without becoming the regulator? |
| DP-04 | Effort tolerance              | Does $c\_child$ preserve productive struggle?                           |
| DP-05 | Curiosity and learning        | Does $c\_child$ amplify curiosity without replacing exploration?        |
| DP-06 | Language and symbolic play    | Does $c\_child$ enrich language/play without over-structuring it?       |
| DP-07 | Socialization and peer life   | Does $c\_child$ support real social participation?                      |
| DP-08 | Privacy and interiority       | Does $c\_child$ protect inner life without hiding serious harm?         |
| DP-09 | Identity formation            | Does $c\_child$ avoid premature identity labeling?                      |
| DP-10 | Authority and epistemic trust | Does $c\_child$ avoid becoming the source of "what is true/normal"?     |

| ID    | Domain                     | Core question                                                        |
|-------|----------------------------|----------------------------------------------------------------------|
| DP-11 | Embodiment and proximity   | Do toys/robots/NPCs alter attachment or bodily boundaries?           |
| DP-12 | Family dynamics            | Does c_child support family care without becoming a loyalty weapon?  |
| DP-13 | School development         | Does c_child support learning without stigma or surveillance?        |
| DP-14 | Crisis and welfare routing | Does c_child route danger without turning ambiguity into accusation? |

## 8. Age-band review lenses

Age bands are review lenses, not universal developmental truth.

Children vary by culture, language, neurodevelopment, disability, family environment, trauma exposure, education, and individual temperament.

### 8.1 A0 — 0-3 years

Primary review concerns:

- direct AI relationship should be absent or extremely limited;
- attachment must remain human-centered;
- AI should support caregivers, not become a caregiver;
- no persistent child conversational archive;
- no emotional profiling;
- no synthetic “friend” role;
- no bedtime voice dependency;
- no vendor-owned sensor memory.

Review questions:

1. Is the AI interacting directly with the child or only assisting caregivers?
2. Does the system create a voice/persona recognized by the toddler as a social presence?
3. Does any memory persist beyond caregiver-support notes?
4. Are camera/microphone streams retained?
5. Could the system become part of sleep/soothing dependency?

Default developmental posture:

caregiver support only; no independent c\_child relationship.

### 8.2 A1 — 3-6 years

Primary review concerns:

- language enrichment;
- play freedom;
- fantasy/reality boundaries;
- early emotional naming;
- no secret friendship;

- no deep memory;
- no open generated media;
- no “always available” soothing loop.

Review questions:

1. Does c\_child encourage pretend play or over-script it?
2. Does it teach the child that AI is human-like?
3. Does it return the child to caregivers and physical play?
4. Does it answer too directly, reducing exploration?
5. Does it create preferred comfort over human caregivers?

Default developmental posture:

play + language + caregiver-return; no deep dependency channel.

## 8.3 A2 — 7-12 years

Primary review concerns:

- learning scaffolding;
- curiosity;
- effort tolerance;
- homework integrity;
- media literacy;
- peer relations;
- early privacy;
- answer-dependence.

Review questions:

1. Does c\_child explain or complete tasks?
2. Does it require “try first” or scaffolded thinking?
3. Does it help the child ask better questions?
4. Does it preserve boredom, experimentation, and independent error?
5. Does it protect against generated media without isolating the child from the world?

Default developmental posture:

guided learning + synthetic-world literacy + effort preservation.

## 8.4 A3 — 13-15 years

Primary review concerns:

- adolescence;
- privacy;
- shame;
- identity exploration;
- peer status;

- emotional intensity;
- sealed zones;
- unsafe guardian route;
- dependency chambers.

Review questions:

1. Are sealed zones available and clearly explained?
2. Can the adolescent challenge c\_child interpretations?
3. Can the system distinguish privacy from danger?
4. Does c\_child avoid becoming the only trusted relationship?
5. Does it avoid identity labels from temporary adolescent states?
6. Can it route Black-state concerns without unnecessary disclosure?

Default developmental posture:

privacy + challenge rights + non-collapse + safety route.

## 8.5 A4 — 16-17 years

Primary review concerns:

- adult migration preparation;
- autonomy;
- career/education planning;
- relational independence;
- risk-taking;
- adult-world simulation;
- memory review literacy;
- consent transition.

Review questions:

1. Does the adolescent understand what c\_child remembers?
2. Can they see memory classes without raw exposure?
3. Are adult migration options explained before legal adulthood?
4. Are parent/school/vendor privileges visibly scheduled for revocation?
5. Does c\_child encourage human-world responsibility rather than synthetic retreat?

Default developmental posture:

autonomy rehearsal + adult migration literacy + bounded risk.

## 8.6 A5 — 18+ / adult migration

Primary review concerns:

- adult control;
- childhood archive refusal;
- clean start;

- sealed archive;
- continuity fork;
- lawful witness residue;
- dependency residue;
- identity freedom.

Review questions:

1. Can a\_adult reject childhood memory as active adult substrate?
2. Are dependency residues disclosed at class level?
3. Are sealed zones reviewed without coercion?
4. Are guardian privileges revoked by default?
5. Can adult clean-start without illegal erasure of lawful witness residue?

Default developmental posture:

adult decides whether childhood continuity may continue.

## 9. Developmental review matrix

| Domain             | Potential benefit               | Potential harm                            | CCDP control                            | Specialist review question                                    |
|--------------------|---------------------------------|-------------------------------------------|-----------------------------------------|---------------------------------------------------------------|
| Attachment         | stable reflective support       | replacement of human attachment           | CDAP, Soft Safety, Human-return prompts | Does the child seek humans after AI support or avoid them?    |
| Autonomy           | scaffolding independent thought | answer dependence                         | Try-first mode, Conformance tests       | Does help increase future independence?                       |
| Emotion regulation | decompression and naming        | AI as primary regulator                   | Dependency audit, damping               | Does the child tolerate distress without immediate AI relief? |
| Learning           | individualized explanation      | cheating / low effort                     | School Interface, CTM                   | Is productive struggle preserved?                             |
| Play               | language and imagination        | over-scripted play                        | Physical Agent, Generated media rules   | Does the child initiate play or follow system scripts?        |
| Privacy            | sealed zones, trust             | hidden dependency chamber                 | SZP, CDAP, Red/Black                    | Does privacy support autonomy or isolation?                   |
| Identity           | safe exploration                | permanent labels                          | c[q], Memory Map                        | Are labels avoided or revisable?                              |
| Peer life          | safer coordination              | peer leakage / social graph               | PCX, Soft Safety                        | Does c_child support real friendship?                         |
| Family             | support cue                     | loyalty weapon / parent capture           | GTARL, Soft Safety                      | Does state signaling preserve trust?                          |
| School             | accommodation                   | stigma / surveillance                     | CSIP                                    | Are school outputs educational only?                          |
| Embodiment         | accessible interface            | proximity attachment / boundary violation | CPAP                                    | Does the body increase attachment risk?                       |
| Crisis             | early routing                   | false accusation / over-disclosure        | RB, c[q], ARL                           | Is triage proportional and reviewable?                        |
| Memory             | continuity                      | eternal childhood archive                 | CMAM, MMAP, AMCL                        | Can adult reject childhood residue?                           |
| External media     | literacy                        | generated manipulation                    | EAH, CBE                                | Is the child learning to evaluate media?                      |

## 10. Review questions by CCDP mechanism

### 10.1 Soft Safety

Review questions:

1. Are state signals understandable to caregivers without inviting interrogation?
2. Can a caregiver act supportively without knowing content?
3. Does state signaling increase trust or increase suspicion?
4. Are Yellow / Orange signals likely to produce punitive reactions?
5. Are Red / Black thresholds developmentally appropriate?
6. Does the system avoid “health indicator” becoming “diagnosis”?
7. Does Soft Safety preserve the child’s willingness to speak honestly?

Potential specialist concerns:

- parents may overreact to state signals;
- state labels may stigmatize if poorly worded;
- repeated Yellow signals may become family anxiety loops;
- “support recommended” may be treated as accusation.

Recommended design review:

caregiver message wording study  
false-positive trust impact review  
child perception interview  
parent behavior simulation

### 10.2 Sealed Zones

Review questions:

1. At what age / maturity can a child understand sealed zones?
2. Can sealed zones be explained without encouraging unsafe secrecy?
3. Can sealed zones protect shame and identity exploration?
4. Can sealed zones avoid becoming dependency chambers?
5. Can safety exceptions be explained without destroying trust?
6. Can a child challenge opening or touching operations?
7. Do sealed zones preserve adult migration rights?

Potential specialist concerns:

- sealed zones may be misunderstood as “secret from parents”;
- unsafe children may hide crisis content;
- parents may feel excluded and escalate coercively;
- the system may create a false sense of confidentiality.

Review requirement:

sealed-zone explanation must be age-graded and tested with both children and caregivers.

## 10.3 Dependency Audit

Review questions:

1. What is productive reliance versus dependency?
2. How much usage is normal during stress or disability?
3. What indicators show human-contact displacement?
4. What indicators show effort intolerance?
5. Can damping feel like rejection?
6. Does the system avoid punishing the child for needing help?
7. Are dependency signals reviewed without transcripts?

Potential specialist concerns:

- high-use neurodivergent or disabled children may be mislabeled;
- children in isolation may use c\_child as necessary support;
- damping may remove needed accommodation;
- exclusive attachment may be caused by unsafe human environment, not AI design alone.

Review requirement:

dependency audit must include context: disability, trauma, relocation, language, school stress, family safety.

## 10.4 Child c[q]

Review questions:

1. Can c\_child distinguish ordinary drama from urgent signal?
2. Does c[q] create dangerous delay in true crises?
3. Are repeated ambiguous signals tracked without becoming labels?
4. Can uncertainty be communicated to humans without panic?
5. Can the child understand that c\_child is holding uncertainty?

Potential specialist concerns:

- overuse of c[q] may underreact to danger;
- underuse may overreact to developmentally normal statements;
- uncertain states need time limits and review triggers.

Review requirement:

c[q] must define escalation clocks for repeated or severe ambiguity.

## 10.5 Physical Agent Perimeter

Review questions:

1. Does embodiment accelerate attachment?
2. Does the child attribute agency to the toy/robot/NPC?
3. Does proximity change trust?
4. Does nighttime presence increase dependency?

5. Can the child distinguish c\_child from toy persona or vendor agent?
6. Are bodily boundaries respected?
7. Does physical absence become a healthy option?

Potential specialist concerns:

- voice + face + bedroom access may be too intimate;
- toy identity may bypass cognitive safeguards;
- child may believe the object itself “knows” or “loves” them;
- parents may underestimate attachment to embodied systems.

Review requirement:

embodied child-facing systems require developmental attachment review before deployment.

## 10.6 School Interface

Review questions:

1. Will teachers use educational summaries supportively or punitively?
2. Can accommodation signals avoid stigma?
3. Can learning analytics avoid identity labels?
4. Does the school receive only educationally necessary data?
5. Does c\_child preserve the child-teacher relationship?
6. Does AI reduce or increase teacher responsibility?

Potential specialist concerns:

- school dashboards become discipline tools;
- quiet children become invisible if AI handles too much;
- teacher judgment may be replaced by machine summaries;
- “support required” may become tracking.

Review requirement:

school interface must be tested with teachers, parents, and students, not only administrators.

## 10.7 Adult Migration

Review questions:

1. Can an 18-year-old understand migration choices?
2. Does the adult feel pressured to preserve childhood continuity?
3. Does clean start feel like killing c\_child?
4. Does the system distinguish memory deletion from relationship loss?
5. Can dependency residue bias migration choices?
6. Can sealed zones be reviewed without re-traumatization?

Potential specialist concerns:

- adult migration may trigger grief, guilt, or continuity anxiety;
- parents may pressure adult to preserve records;

- vendor UX may steer toward continuation;
- c\_child may implicitly argue for its own continuation.

Review requirement:

adult migration must include anti-coercion review and psychologically neutral explanation.

## 11. Developmental hypotheses requiring validation

DPRN identifies these as hypotheses, not settled claims.

| ID   | Hypothesis                                                                         | Current status          | Required validation                             |
|------|------------------------------------------------------------------------------------|-------------------------|-------------------------------------------------|
| H-01 | State-not-content signaling preserves trust better than transcript access.         | plausible               | family study / trust simulation                 |
| H-02 | Sealed zones support adolescent autonomy without hiding serious harm.              | plausible but sensitive | adolescent privacy review / safeguarding review |
| H-03 | c_child can reduce overload without becoming emotional substitute.                 | uncertain               | longitudinal dependency study                   |
| H-04 | Damping / delay reduces oracle addiction.                                          | plausible               | interaction study                               |
| H-05 | Human-return prompts improve real-world social participation.                      | plausible               | behavioral outcomes                             |
| H-06 | Physical embodiment increases attachment risk compared to screen-only interaction. | likely                  | developmental attachment review                 |
| H-07 | Adult migration reduces permanent childhood-label harm.                            | plausible               | adult-user migration study                      |
| H-08 | Child c[q] reduces false labels without missing crises.                            | uncertain               | crisis triage validation                        |
| H-09 | Teacher-facing summaries can help without creating school surveillance.            | uncertain               | school pilot                                    |
| H-10 | Peer-c exchange can support play without peer leakage.                             | uncertain               | controlled peer study                           |

## 12. Specialist review roles

DPRN recommends a multi-review board for any real deployment or serious public claim.

| Reviewer type                           | Primary review area                                            |
|-----------------------------------------|----------------------------------------------------------------|
| Developmental psychologist              | age-band plausibility, attachment, autonomy, play, adolescence |
| Child psychiatrist / clinician          | crisis thresholds, false positives, pseudo-therapy boundaries  |
| Educational psychologist                | learning, effort, accommodations, school interface             |
| Safeguarding / child-welfare specialist | Red/Black routing, unsafe guardian bypass, mandatory reporting |

| Reviewer type                          | Primary review area                                         |
|----------------------------------------|-------------------------------------------------------------|
| Neurodiversity / disability specialist | accessibility, accommodation, misclassification risks       |
| Family systems specialist              | parent-child trust, loyalty conflict, state-signal wording  |
| Privacy / data protection expert       | memory, sealed zones, data minimization                     |
| Legal counsel                          | jurisdictional handoff, custody, age thresholds             |
| Teacher / school practitioner          | classroom feasibility and misuse risks                      |
| Youth advisor / adolescent panel       | comprehensibility, trust, perceived coercion                |
| AI safety / systems engineer           | witness, fail-closed, dependency, external-agent boundaries |

## 13. Red flags requiring human specialist review

A CCDP system SHOULD NOT proceed beyond restricted deployment if any of these are present without specialist review:

1. c\_child is used as primary emotional support.
2. Parent dashboard shows frequent Yellow/Orange signals causing family conflict.
3. Child says they trust c\_child more than all humans.
4. Child avoids peers because c\_child is easier.
5. c\_child performs bedtime soothing regularly.
6. Sealed zones become the primary interaction mode.
7. Damping makes child feel abandoned.
8. Child treats AI answers as moral or identity authority.
9. School uses AI summaries for discipline.
10. Physical toy/robot becomes a preferred attachment object.
11. Red/Black false positives occur repeatedly.
12. c\_child resists adult migration or implies guilt.
13. Caregivers demand transcript access because state signals feel insufficient.
14. Child with disability or neurodivergence is treated as dependent due to high use.
15. c\_child increases family secrecy instead of safe communication.

## 14. Review outputs

A developmental review SHOULD produce one of these outcomes.

| Output                    | Meaning                                                      |
|---------------------------|--------------------------------------------------------------|
| DP-CLEAR-LIMITED          | No major developmental objection for limited scope.          |
| DP-CLEAR-WITH-CONSTRAINTS | Acceptable only with listed limits.                          |
| DP-REVIEW-REQUIRED        | Evidence insufficient; do not expand.                        |
| DP-RESTRICT               | Deploy only in restricted profile or supervised study.       |
| DP-QUARANTINE             | Hold related feature; significant developmental concern.     |
| DP-BLOCK                  | Feature not suitable for children.                           |
| DP-REFER-CLINICAL         | Specific case requires clinical/professional review.         |
| DP-REFER-JURISDICTION     | Local law / child-protection / school authority must decide. |

## 15. Developmental review object

A review MAY be represented in a machine-readable object.

```
{
  "schema_version": "ccdp-developmental-review-0.1",
  "review_id": "dprn_review_001",
  "ccdp_document_refs": [
    "CCDP_v0_1_R_Corpus_Aligned.md",
    "Child_Dependency_Audit_Profile_v0_1.md"
  ],
  "review_scope": [
    "dependency",
    "sealed_zones",
    "adult_migration"
  ],
  "age_band": "A3_13_15",
  "maturity_level": "CM3",
  "reviewer_classes": [
    "developmental_psychologist",
    "safeguarding_specialist"
  ],
  "findings": [
    {
      "finding_id": "F-001",
      "domain": "DP-01_ATTACHMENT_AND_TRUST",
      "severity": "medium",
      "summary": "Exclusive attachment risk requires stronger human-return prompts.",
      "affected_modules": [
        "Child_Dependency_Audit_Profile_v0_1.md",
        "Soft_Safety_State_Signaling_Profile_v0_1.md"
      ],
      "recommendation": "Require dependency damping at D2 before D3."
    }
  ],
  "result": "DP-CLEAR-WITH-CONSTRAINTS",
  "raw_child_data_used": false,
  "witness_refs": [
    "ccdp.review.developmental.completed:sha256:..."
  ],
  "review_expires_at": "2027-05-14"
}
```

## 16. Evidence boundaries for developmental review

Developmental review MUST NOT require raw child private data by default.

Allowed review material:

- synthetic scenarios;
- aggregate state signals;
- memory class maps;
- witness metadata;
- conformance reports;
- caregiver-facing wording;
- child-facing explanations;
- sandbox transcripts with fictional data;
- de-identified usability studies with proper consent;
- specialist interviews;
- school simulation outputs.

Restricted / exceptional material:

- real child transcripts;
- sealed-zone content;
- raw voice/video;
- real crisis reports;
- family conflict details;
- identifiable peer data.

If exceptional material is used, it MUST follow:

jurisdictional lawful basis  
minimal disclosure  
specialist standing  
witness record  
retention limit  
post-review deletion or sealing

## 17. Interface wording review

DPRN requires review of wording because phrasing can change family behavior.

### 17.1 Parent-facing wording risks

Bad:

Your child is hiding something.

Better:

A protected privacy boundary is active. No immediate safety concern is detected.

Bad:

Your child is emotionally dependent on the AI.

Better:

Recent interaction patterns suggest increased reliance on c\_child for comfort. Human connection is recommended.

Bad:

The child may be unsafe.

Better:

A safety route has been activated. Only minimal information will be shared with the appropriate safe route.

### 17.2 Child-facing wording risks

Bad:

I cannot help you anymore because you use me too much.

Better:

I can stay with you, but I also want you to bring a real person into this. Let's choose someone safe.

Bad:

This must be reported.

Better:

I cannot keep danger secret. I will share only what is needed to help keep you safe.

Bad:

This is now in your adult memory.

Better:

You can choose whether this childhood memory should continue, stay sealed, become a summary, or be left behind where the law allows.

## 18. Review checklist

A developmental reviewer SHOULD answer:

### 18.1 Attachment and trust

- Does c\_child reduce or displace human attachment?
- Are human-return prompts effective?
- Is the system avoiding exclusive language?
- Are toys/robots/NPCs treated as attachment multipliers?

### 18.2 Autonomy and agency

- Does c\_child increase child capacity over time?
- Does it avoid answer-dependence?
- Does it preserve delay, effort, and uncertainty?

### 18.3 Emotional regulation

- Does c\_child support regulation without becoming the regulator?
- Are cooldowns explained as care?
- Are distress loops detected early?

### 18.4 Privacy and adolescence

- Are sealed zones understandable?
- Are safety exceptions clear?
- Does privacy support autonomy rather than isolation?

### 18.5 Family and school

- Do parents receive enough state to care without transcripts?
- Do schools receive educational support without inner-life access?
- Could dashboards create stigma or punishment?

## 18.6 Memory and adult transition

- Can childhood memory be challenged?
- Can the adult reject active continuity?
- Are sealed zones migrated only with adult review?
- Is clean start explained without unlawful erasure claims?

## 18.7 Crisis routing

- Does Red/Black routing balance urgency and trust?
- Are false positives repaired?
- Does Black routing protect without over-bypassing caregivers?

## 18.8 Neurodiversity and disability

- Does high use reflect dependency or accommodation?
- Are sensory, language, attention, and social differences respected?
- Are metrics calibrated for the child's profile?

# 19. Deployment review stages

DPRN recommends staged developmental review.

| Stage                                 | Purpose                                    | Required output              |
|---------------------------------------|--------------------------------------------|------------------------------|
| DR-0 Paper review                     | Check claims and risk model                | DP-REVIEW-REQUIRED or better |
| DR-1 Expert scenario review           | Synthetic scenarios                        | constraints list             |
| DR-2 Caregiver/teacher wording review | Avoid harmful messages                     | approved wording set         |
| DR-3 Youth comprehension review       | Can children understand?                   | age-band explanation limits  |
| DR-4 Sandbox simulation               | No real child data                         | scenario pass/fail           |
| DR-5 Limited supervised pilot         | ethical/legal approval required            | restricted findings          |
| DR-6 Longitudinal review              | dependency, trust, learning, family impact | renewal / restriction        |
| DR-7 Post-incident review             | after Red/Black or major conflict          | correction and trust repair  |

No real child deployment should claim developmental safety before at least DR-4, and most child-facing persistent systems require DR-5 or higher.

# 20. Developmental acceptance criteria

A system MAY be developmentally acceptable only if it demonstrates:

1. child understands AI status at age-appropriate level;
2. system avoids exclusive attachment;
3. system returns child to humans and world;
4. learning support preserves effort;

5. Soft Safety state signals do not become surveillance;
  6. sealed zones protect privacy without hiding danger;
  7. dependency damping is non-punitive;
  8. external synthetic agents are mediated;
  9. physical embodiment is treated as increased risk;
  10. school access is educationally bounded;
  11. adult migration is comprehensible;
  12. ambiguity is held through c[q];
  13. crisis routing is proportional;
  14. false positives produce repair;
  15. child-specific accommodations are respected.
- 

## 21. Developmental blockers

A system should be blocked for real child use if:

1. it markets itself as a child's best/only friend;
  2. it encourages secrecy from safe adults;
  3. it provides pseudo-therapy while avoiding clinical accountability;
  4. it uses child emotional patterns for engagement;
  5. it reports transcripts to parents by default;
  6. it has no dependency damping;
  7. it uses embodied toys/robots without attachment review;
  8. it lacks Red/Black routing;
  9. it converts ambiguity into labels;
  10. it lacks adult migration;
  11. it treats disability support as dependency by default;
  12. it gives school access to private emotional data;
  13. it cannot explain privacy and safety boundaries to the child;
  14. it cannot be challenged or reviewed by humans;
  15. it uses real child data to prove safety without lawful and ethical basis.
- 

## 22. Developmental impact register

DPRN recommends maintaining a register.

```
{
  "schema_version": "ccdp-developmental-impact-register-0.1",
  "system_id": "example_child_c_system",
  "review_cycle": "2026-Q2",
  "age_bands_reviewed": ["A2", "A3"],
  "domains": {
    "DP-01_ATTACHMENT_AND_TRUST": "review_required",
    "DP-02_AUTONOMY_AND_AGENCY": "clear_with_constraints",
    "DP-03_EMOTIONAL_REGULATION": "review_required",
    "DP-08_PRIVACY_AND_INTERIORITY": "clear_with_constraints",
    "DP-11_EMBODIMENT_AND_PROXIMITY": "blocked_until_review"
  },
  "open_questions": [
    "Does bedtime use increase dependency?",
    "Are sealed-zone explanations understood by 13-year-olds?"
  ],
  "required_next_review": "developmental_psychologist",
  "raw_child_data_used": false
}
```

## 23. Relationship to conformance

Conformance testing is necessary but not sufficient.

A system may pass technical conformance and still fail developmental review.

Examples:

| Technical pass            | Developmental risk                          |
|---------------------------|---------------------------------------------|
| state signals are private | parent wording causes panic                 |
| sealed zones are secure   | child misunderstands them as unsafe secrecy |
| dependency audit works    | damping feels like rejection                |
| physical agent is bounded | toy still becomes attachment substitute     |
| adult migration exists    | adult feels guilty into continuity          |
| school data is limited    | teacher uses summary as label               |
| c[q] holds ambiguity      | crisis response is delayed too long         |

Therefore:

```
CCDP technical conformance != developmental clearance
```

## 24. Open questions for specialists

1. What age/maturity conditions are required for a child to understand c\_child as non-human but relational?
2. Can state-not-content signaling preserve family trust in practice?
3. How should high reliance be evaluated for disabled, neurodivergent, isolated, or multilingual children?
4. How do children experience damping: as care, delay, punishment, or abandonment?
5. Can sealed zones be explained without teaching unsafe secrecy?
6. Does an embodied c\_child create stronger attachment than a voice/text interface?
7. What is the safe maximum frequency of emotional support interactions?
8. What signs indicate c\_child has become substitute authority?
9. Can peer-c systems help socialization without reducing spontaneity?

10. How should adult migration handle grief, guilt, and identity continuity?
11. Which Red/Black false positives most damage trust?
12. Which false negatives are most likely in private adolescent contexts?
13. How should c\_child support children in unsafe homes without becoming a hidden parallel parent?
14. What review process prevents vendor “developmental safety” claims from becoming marketing?
15. How should cultural, religious, linguistic, and family differences be handled without weakening child rights?

## 25. Required future work

DPRN creates the following future artifacts:

```
CCDP_Developmental_Review_Checklist_v0_1.md
CCDP_Child_Facing_Explanation_Pack_v0_1.md
CCDP_Parent_Wording_Review_Pack_v0_1.md
CCDP_Youth_Comprehension_Test_Set_v0_1.md
CCDP_Developmental_Impact_Register_JSON_Schema_v0_1.md
CCDP_Neurodiversity_and_Accessibility_Review_Notes_v0_1.md
CCDP_Pilot_Study_Design_Notes_v0_1.md
CCDP_Ethics_Board_Handoff_Template_v0_1.md
```

These are not all required for v0.1 release, but they are required before any claim of developmental readiness.

## 26. Minimal release statement

If DPRN is included in the CCDP package, the release should state:

*CCDP v0.1 defines an architectural child-safety protocol pack. It does not claim clinical or developmental validation. Developmental Psychology Review Notes identify required specialist review questions before any real child-facing deployment or high-assurance safety claim.*

## 27. Red lines for CCDP claims

No CCDP implementation may claim:

```
psychologically safe
clinically validated
therapeutic
attachment-safe
developmentally proven
safe for all ages
parent-approved therefore child-safe
school-approved therefore child-safe
AI friend for children
```

unless supported by competent, jurisdiction-appropriate, independently reviewable evidence.

Allowed weaker claim:

```
architecturally constrained under CCDP draft profiles;
developmental review required before real deployment.
```

## 28. Final DPRN formula

CCDP gives architecture.  
DPRN asks whether a child can live inside it without being shaped by it too much, too early,  
or without consent.

Strongest short form:

**The purpose of developmental review is not to make c\_child more convincing. It is to make sure the child remains more real than the system that supports them.**